Module 7 :: Registers and Counters

Practice Problems

1. Draw the schematic diagram of a 4-bit shift register using: (i) S-R flip-flops, (ii) J-K flip-flops, (iii) D flip-flops.
2. Draw the schematic diagram of a 4-bit parallel-in parallel-out (PIPO) register using D flip-flops.
3. Draw the schematic diagram of a 4-bit ring counter, and draw the state table in response to applied clock pulses. Assume that the register is initialized to 1000. After how many cycles do the states repeat?
4. Draw the schematic diagram of a 4-bit twisted ring or Johnson counter, and draw the state table in response to applied clock pulses. Assume that the register is initialized to 0000. After how many cycles do the states repeat?
5. Draw the schematic diagram of a modulo-16 binary ripple counter using T flip-flops. If a 100 MHz clock is applied to the counter in, what would be frequencies of the signals generated in the different flip-flop outputs?
6. Repeat (5) using J-K flip-flop.
7. A modulo-M counter and a modulo-N counter are connected in cascade. The combination is a counter that counts modulo-K. What is the value of K?
8. Draw the schematic diagram of a 4-bit binary ripple counter that counts: (i) modulo-10, (ii) modulo 12, (iii) modulo 14.
9. How can you design a modulo-1000 counter, using BCD counters as the basic building blocks? [Note that BCD counters and modulo-10 counters are the same thing]
10. Using modulo-10 binary counters as basic building blocks, design a digital clock that produces BCD outputs corresponding to hour, minute and second.
11. For an n-bit LFSR, can all the 2^n distinct states lie in a cycle? Justify your answer.
12. An n-bit LFSR is said to be a maximal-length LFSR if the number of distinct states in a cycle is $2^n - 1$. If the number of inputs to the exclusive-OR gate is odd, justify that the LFSR can never be maximal-length.
13. Design a circuit that can compute the N-th Fibonacci number, given the value of N.